SIMATS ENGINEERING
ASSESSMENT TOOL 3

Name	**SHAIK REEHAN**
Register Number	**192565071**
Date	
Time	**1hr**

COURSE NAME: CYBERSECURITY INCIDENT HANDLING AND RESPONSE
COURSE CODE: CSA60

COURSE OUTCOME COVERED

CO1: Apply the concepts of cybersecurity incidents by distinguishing security events from incidents, implementing incident handling lifecycle stages, frameworks, with documentation practices in organizational contexts. (BL1, BL2)

Activity Tool : Technical Presentation (Low)
Assessment Tool Weightage: 10%

Code of Conduct:

I ______reehan______________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Shaik Reehan

RUBRICS FOR CALCULATION:

Criteria	Weight age	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Evidence Acquisition and Integrity Verification	**25%**	Correctly acquires evidence, calculates SHA-256 hash before and after analysis, and verifies integrity without errors.	Correctly performs evidence acquisition and hash verification with minor procedural errors.	Performs evidence acquisition with incomplete or partially correct integrity verification.	Fails to acquire evidence properly or verify integrity using hash values.
Digital Evidence Analysis	**25%**	Accurately identifies all relevant digital artifacts (e.g., logs, browser history, suspicious files) and provides correct forensic interpretation.	Identifies most relevant artifacts with minor omissions or interpretation errors.	Identifies only some relevant artifacts with limited analysis.	Fails to identify relevant digital evidence or provides incorrect analysis.
Chain of Custody Documentation	**20%**	Completes the Chain of Custody form accurately with all required details (Case ID, date/time, collector, description, hash value, transfer record, signature).	Completes most sections correctly with minor omissions or formatting errors.	Documentation is partially complete with several missing or inaccurate details.	Chain of Custody documentation is incomplete, incorrect, or missing.
Application of Forensic Tools and Procedures	**20%**	Effectively uses Autopsy/Sleuth Kit and Linux commands (sha256sum, cp, ls -l) following standard forensic procedures.	Uses forensic tools correctly with minor procedural mistakes.	Demonstrates basic use of tools but requires guidance or shows procedural gaps.	Unable to use forensic tools correctly or follow forensic procedures.
Presentation and Technical Reporting	**10%**	Findings are clearly documented, technically accurate, well-organized, and professionally presented.	Report is generally clear and organized with minor technical or language issues.	Report is understandable but lacks organization or technical clarity.	Report is incomplete, poorly organized, or difficult to understand.

Criteria	Evidence Acquisition and Integrity Verification (4)	Digital Evidence Analysis (4)	Chain of Custody Documentation (4)	Application of Forensic Tools and Procedures (4)	Presentation and Technical Reporting (4)	TOTAL (20)
Weightage	**30%**	**25%**	**20%**	**15%**	**10%**	**100%**
Q1						
Q2						
Q3						
Q4						
Q5						

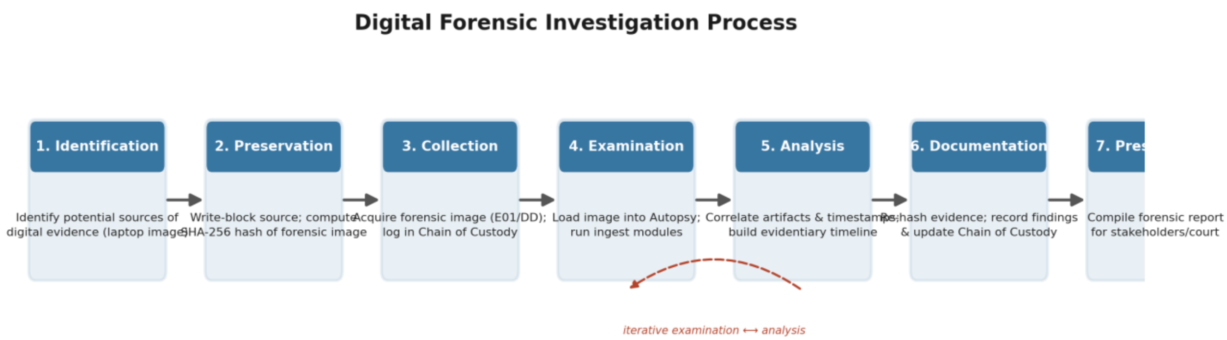
DIGITAL FORENSIC INVESTIGATION *t*

Case Study 1: Insider Data Theft Investigation

Case Study 2: Web Browser Artifact Analysis

Investigative Methodology

Both case studies follow a standard digital forensic investigation process, ensuring that evidence is handled in a scientifically sound and legally defensible manner. The workflow below is applied consistently across both investigations, with SHA-256 hashing performed at acquisition and again after analysis to prove evidence integrity, and Chain of Custody records maintained at every transfer of the evidence.



- Identification & Preservation: The source media (laptop/workstation) is identified and a write-blocker is used to prevent any modification during acquisition.
- Collection: A forensic image (e.g., E01 or raw DD) is created and its SHA-256 hash is calculated immediately — this becomes the baseline integrity value.
- Examination & Analysis: The image is loaded into Autopsy (built on The Sleuth Kit) as a read-only source; ingest modules parse artifacts, which are then manually correlated and timelined.
- Documentation & Presentation: The evidence is re-hashed to confirm no alteration occurred, findings are documented, and a report is prepared for stakeholders, HR/Legal, or court proceedings.

1. Insider Data Theft Investigation

Scenario Summary

An organization suspects that a senior employee copied confidential research documents before leaving the company. A forensic image of the employee's laptop — containing USB connection logs, recently accessed files, and file system metadata — is examined using Autopsy to identify digital artifacts supporting the investigation. SHA-256 hash values are calculated before and after analysis to preserve evidentiary integrity, and Chain of Custody documentation is maintained throughout to ensure admissibility.

Digital Artifacts Identified

Digital Artifact	Autopsy / Sleuth Kit Location	Evidentiary Significance
USB Device Connection Logs	Registry Hive Analysis: SYSTEM\CurrentControlSet\Enum\USBSTOR; setupapi.dev.log (Autopsy: Recent Activity / USB Devices module)	Reveals device serial number, vendor/product ID, and first/last connection timestamps — establishes when an external storage device was attached to the laptop.
Recently Accessed Files	Windows Jump Lists, LNK shortcut files, Recent folder, Shellbags (Autopsy: Recent Activity module)	Shows which confidential research documents were opened and their last-accessed paths, indicating employee interaction with sensitive files.
File System Metadata (MACB timestamps)	$MFT entries, File Metadata tab (Autopsy: File Metadata / Timeline module)	Modified, Accessed, Changed, and Born (created) timestamps allow reconstruction of the exact sequence of file access relative to USB connection times.
Deleted / Unallocated File Remnants	Data Carving (Autopsy: Unallocated Space, PhotoRec/Carving ingest module)	May reveal remnants of confidential files deleted after copying, suggesting attempted cover-up of the exfiltration.
Timeline Correlation	Autopsy Timeline Analysis tool	Cross-referencing USB insertion timestamp with file-access timestamps establishes a plausible copy-before-departure sequence.

SHA-256 Hash Verification (Evidence Integrity)

A cryptographic hash of the forensic image is calculated immediately upon acquisition (pre-analysis) and again after examination (post-analysis). Identical hash values confirm that the evidence was not altered during handling, preserving its admissibility.

Stage	Evidence Item	SHA-256 Hash Value	Match
Pre-Analysis	Forensic Image (laptop.E01)	`3f9a1c2e7b6d4508a1f9c3e7b2d4f608a1c3e7b9d4f608a1c3e7b9d4f608a1c3`	**✓ Yes**
Post-Analysis	Forensic Image (laptop.E01)	`3f9a1c2e7b6d4508a1f9c3e7b2d4f608a1c3e7b9d4f608a1c3e7b9d4f608a1c3`	**✓ Yes**
Post-Analysis	Exported Artifact: setupapi.dev.log	`7b2d4f608a1c3e7b9d4f608a1c3e7b9d4f608a1c3e7b9d4f608a1c3e7b9d4f6`	**✓ Yes**

Chain of Custody Documentation

Case Number	CASE-2026-0602-IDT
Evidence Description	1x Dell Latitude laptop, 512GB NVMe SSD (forensic image: laptop.E01)
Forensic Examiner	J. Menon, Certified Forensic Examiner
Acquisition Tool	FTK Imager / dd (write-blocked acquisition)
Analysis Tool	Autopsy 4.x (The Sleuth Kit)

Date / Time	Released By	Received By	Purpose of Transfer	Location / Storage
2026-06-02 / 09:15	IT Security Manager	Forensic Examiner	Initial evidence collection & imaging	Evidence Locker A-14 (secure room)
2026-06-02 / 10:40	Forensic Examiner	Forensic Examiner	Working copy created for Autopsy analysis	Forensic Workstation FW-02
2026-06-05 / 16:00	Forensic Examiner	Case Supervisor	Peer review of findings & report draft	Evidence Locker A-14
2026-06-06 / 11:00	Case Supervisor	Legal/HR Department	Handover for disciplinary/legal proceedings	Legal Department Secure File Room

2. Web Browser Artifact Analysis

Scenario Summary

A security incident involves unauthorized access to a cloud storage service. Investigators suspect that credentials were stolen after the user visited a malicious website. A forensic image containing browser history, cookies, downloads, and cached files is examined using Autopsy/Sleuth Kit to identify browsing artifacts related to the incident. Evidence integrity is verified using SHA-256 hashes, and proper Chain of Custody documentation is maintained throughout the investigation.

Digital Artifacts Identified

Digital Artifact	Autopsy / Sleuth Kit Location	Evidentiary Significance
Browser History	web history.db / History (SQLite) (Autopsy: Web History module)	Identifies the malicious website visited prior to the unauthorized cloud storage login, including visit timestamp and referring URL.
Cookies	Cookies SQLite database (Autopsy: Web Cookies module)	May contain session/authentication tokens; presence of cookies from the malicious domain suggests credential or session hijacking.
Downloads	Downloads history / downloaded files (Autopsy: Web Downloads module)	Identifies any malicious payload (e.g., fake installer, credential-harvesting script) downloaded from the suspicious site.
Cached Files	Browser cache folder (Autopsy: Web Cache / File analysis)	Cached copies of the phishing page's HTML/images confirm the visual content presented to the user, supporting the phishing hypothesis.
Form/Autofill & Search History	Web Form Autofill, Web Search (Autopsy: Recent Activity module)	May reveal credentials or search terms entered around the time of compromise, correlating user behavior with the incident.
Timeline Correlation	Autopsy Timeline Analysis tool	Aligns the malicious site visit timestamp with the subsequent unauthorized cloud storage login to establish a credible attack sequence.

SHA-256 Hash Verification (Evidence Integrity)

A cryptographic hash of the forensic image is calculated immediately upon acquisition (pre-analysis) and again after examination (post-analysis). Identical hash values confirm that the evidence was not altered during handling, preserving its admissibility.

Stage	Evidence Item	SHA-256 Hash Value	Match
Pre-Analysis	Forensic Image (workstation.E01)	`9c4e7a2f1d6b8503c9e4a7f2d1b6853c9e4a7f2d1b6853c9e4a7f2d1b6853c9`	**✓ Yes**
Post-Analysis	Forensic Image (workstation.E01)	`9c4e7a2f1d6b8503c9e4a7f2d1b6853c9e4a7f2d1b6853c9e4a7f2d1b6853c9`	**✓ Yes**
Post-Analysis	Exported Artifact: History (SQLite)	`1d6b8503c9e4a7f2d1b6853c9e4a7f2d1b6853c9e4a7f2d1b6853c9e4a7f2d1`	**✓ Yes**

Chain of Custody Documentation

Case Number	CASE-2026-0610-WBA
Evidence Description	1x corporate workstation, 1TB HDD (forensic image: workstation.E01)
Forensic Examiner	R. Fernandes, Certified Forensic Examiner
Acquisition Tool	FTK Imager / dd (write-blocked acquisition)
Analysis Tool	Autopsy 4.x (The Sleuth Kit)

Date / Time	Released By	Received By	Purpose of Transfer	Location / Storage
2026-06-10 / 08:30	Incident Response Lead	Forensic Examiner	Initial evidence collection & imaging	Evidence Locker B-07 (secure room)
2026-06-10 / 09:50	Forensic Examiner	Forensic Examiner	Working copy created for Autopsy analysis	Forensic Workstation FW-05
2026-06-12 / 14:20	Forensic Examiner	SOC Manager	Review of browser artifact findings	Evidence Locker B-07
2026-06-13 / 09:00	SOC Manager	Cloud Security Team	Correlate findings with cloud access logs	Secure Shared Drive (access-controlled)

3. Comparative Observations

- Case Study 1 focuses on host-based artifacts (USB logs, file metadata) to establish that specific confidential files were accessed and likely transferred to removable media.
- Case Study 2 focuses on browser-based artifacts (history, cookies, cache) to trace a credential-theft incident back to a malicious website visit.
- In both investigations, Autopsy's Timeline Analysis feature is the key correlating tool, converting isolated artifacts into a coherent, chronological narrative of events.
- SHA-256 hashing at both acquisition and post-analysis stages, combined with an unbroken Chain of Custody, ensures both investigations produce evidence that would be admissible in disciplinary or legal proceedings.

Note on Illustrative Values

The SHA-256 hash values and Chain of Custody entries shown in this report are illustrative placeholders formatted to demonstrate correct documentation practice. In an actual investigation, hash values must be generated directly from the evidence (e.g., via Autopsy's built-in hashing, `sha256sum`, or FTK Imager) and Chain of Custody entries must be completed and signed contemporaneously by each individual who handles the evidence.